

HALO IT SERVICES 365

MASTER IMPLEMENTATION SPECIFICATION

FILE 1 OF 5

Executive Overview, Current State, Core Architecture, Dashboard, Ticketing, Knowledge Base, AI Platform

Version: 1.0

Date: 2026-06-09

Status: Master Requirements Specification

Repository: Halo IT Services 365

1. PROJECT PURPOSE

Halo IT Services 365 is intended to become a complete centralized IT Operations, MSP, Security, Automation, Learning, Development, AI and Business Operations Platform.

The purpose is to eliminate dependency on dozens of separate tools and instead provide a single integrated platform capable of:

- IT Service Management
- Ticketing
- RMM
- Monitoring
- AI Operations
- Knowledge Management
- Client Management
- SLA Management
- Security Operations
- User Management
- HR Functions
- Learning & Certification Tracking
- Accounting
- Invoicing
- Quotations
- Shop Management
- Backup Management

- Password Vault
- Development Operations
- Application Building
- VoIP
- Integrations
- Marketing
- Expansion Framework

All systems must be interconnected while maintaining security boundaries.

2. CURRENT REPOSITORY STATUS

Verified Existing Components

Backend

- Node.js
- Express.js
- Sequelize ORM
- SQLite Database
- JWT Authentication
- Role Based Access Control
- Ticket API
- Dashboard API
- Monitoring API
- User API
- Audit API
- AI Provider Framework

Frontend

- Authentication UI
- Dashboard UI
- Ticket UI
- Client UI
- Monitoring UI
- AI UI
- User Management UI

Security

- JWT
- RBAC
- bcrypt
- Helmet
- CORS

Deployment

- Railway Ready
 - Replit Ready
 - Xneelo Documentation Ready
-

3. DEVELOPMENT DIRECTIVE

The existing repository is the foundation.

The system must be expanded.

The system must not be rewritten.

Existing architecture must be preserved wherever possible.

All additions must integrate into:

- Existing Authentication
- Existing RBAC
- Existing Audit Logging
- Existing Dashboard
- Existing AI Framework

No placeholder systems.

No mock systems.

No dummy APIs.

No fake workflows.

Every feature must either:

- Be fully operational

OR

- Have operational setup and integration capability.
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4. CORE PLATFORM ARCHITECTURE

Core Modules

Dashboard Centre

Central operational overview.

Ticket Centre

All support operations.

Monitoring Centre

RMM and Device Operations.

Knowledge Centre

Internal and client documentation.

AI Centre

Unified AI operations.

Client Centre

Customer portal and management.

Accounting Centre

Invoices, quotes and finance.

Security Centre

SOC functionality.

Learning Centre

Training and certifications.

User Centre

Users, HR and personnel.

Vault Centre

Passwords and secrets.

Backup Centre

Backup management.

Integration Centre

Third-party systems.

Dev Console

Development operations.

Expansion Centre

Future feature deployment.

App Builder

Internal software creation.

Marketing Centre

Advertising and campaigns.

VoIP Centre

Communications platform.

5. DASHBOARD CENTRE

Objective

Provide complete operational visibility.

Dashboard Widgets

Tickets

Display:

- Open Tickets
 - Closed Tickets
 - Pending Tickets
 - Awaiting Approval
 - Awaiting Client Response
 - Escalated Tickets
 - SLA Breaches
 - Upcoming Maintenance
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Monitoring

Display:

- Device Health
 - Offline Devices
 - Critical Alerts
 - Warning Alerts
 - Security Alerts
 - Backup Alerts
 - Patch Status
-

SLA

Display:

- SLA Compliance
 - SLA Breaches
 - Response Time
 - Resolution Time
 - Client Rankings
-

Staff

Display:

- Online Agents
- Offline Agents

- Working Tickets
 - Current Utilisation
 - Training Progress
 - Certification Progress
-

Financial

Display:

- Quotes Pending
 - Quotes Approved
 - Draft Invoices
 - Outstanding Invoices
 - Revenue Tracking
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Client Management

Display:

- Active Clients
 - Contract Clients
 - Non Contract Clients
 - New Clients
 - At Risk Clients
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AI Operations

Display:

- AI Usage
 - AI Requests
 - AI Health
 - AI Costs
 - AI Learning Metrics
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Security

Display:

- Failed Logins
- MFA Failures
- Threat Alerts

- Security Score
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Notifications

Display:

- Global Notifications
 - Critical Alerts
 - Approval Requests
 - Escalations
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6. TICKETING CENTRE

Objective

Provide enterprise-grade ticket lifecycle management.

Ticket Types

Support Ticket

Incidents and support requests.

Service Request

User requests.

Change Request

System modifications.

Requires approval workflow.

Project Ticket

Project work items.

Maintenance Ticket

Planned maintenance.

Emergency Ticket

Critical incidents.

System Generated Ticket

Automatically created from:

- Monitoring
 - Security
 - Backup
 - SLA
 - AI
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Ticket Priorities

- Low
 - Medium
 - High
 - Critical
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Ticket States

- New
 - Assigned
 - In Progress
 - Pending
 - Awaiting Approval
 - Awaiting Client
 - Escalated
 - Resolved
 - Closed
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Ticket Ownership

Support:

- Self
- Client
- Agent
- Department

Must support creation on behalf of:

- Clients
 - Staff
 - Managers
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Ticket Approvals

Change Requests

Must support:

- Client Approval
 - Peer Review
 - Manager Approval
-

Self Approval Logic

Business owners may self approve.

Must include:

- Risk Acceptance Checkbox
 - Timestamp
 - Audit Log
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Ticket Permissions

Granular RBAC.

Examples:

- Create
- Edit
- Close
- Delete
- Approve
- Escalate
- Assign

Team Isolation

Must support:

- 1st Line
- 2nd Line
- 3rd Line
- Projects
- Management

Visibility based on role permissions.

Time Tracking

Integrated time management.

Features:

- Start Timer
- Stop Timer
- Pause Timer
- Manual Entry
- Manual Entry Justification

Integration Support:

- Clockify
- Internal Time Engine

Ticket Statistics

Reports:

- Engineer Performance
- Resolution Times
- Client Statistics
- SLA Compliance
- Workload Metrics

Export:

- PDF
 - CSV
 - Excel
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Ticket Audit Logging

Track:

- Creation
 - Assignment
 - Changes
 - Notes
 - Approvals
 - Closures
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Ticket Automation

Automated creation from:

- Monitoring
 - Security
 - Backup
 - AI
 - SLA
-

Onboarding Tickets

Must support:

- New Staff
 - New Clients
 - New Vendors
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Offboarding Tickets

Must support:

- Staff Departures
 - Client Departures
 - Vendor Access Removal
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7. KNOWLEDGE CENTRE

Objective

Become the central intelligence repository.

Knowledge Categories

Internal

Halo IT documentation.

Client

Client-specific documentation.

Vendor

External systems.

Security

Security procedures.

Learning

Training content.

Knowledge Sources

Manual Entry

Staff created.

AI Generated

AI generated articles.

Internet Research

AI-assisted research.

Vendor Documentation

Imported references.

Knowledge Features

Search

Natural language search.

AI Troubleshooting

Example:

"I cannot upload files to a Seagate NAS"

AI must:

- Analyze issue
 - Search KB
 - Search Internet
 - Suggest Fixes
 - Save Knowledge
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Client Documentation

Per client:

- Systems
 - Infrastructure
 - Vendors
 - Contacts
 - Procedures
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Permissions

View/Edit rights by role.

Audit Logging

All modifications logged.

8. AI PLATFORM

Objective

Provide Halo IT's intelligent operational assistant.

Internal AI

Capabilities:

- Chat
- Code Assistance

- Documentation
 - Ticket Assistance
 - Monitoring Analysis
 - Security Analysis
 - Reporting
 - Automation
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AI Providers

Primary:

DeepSeek v4 Pro via Azure Foundry.

Secondary:

- Azure OpenAI
 - OpenAI
 - Ollama
 - OpenRouter
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Model Management

Super Admin can:

- Add Models
 - Remove Models
 - Switch Models
 - Configure Failover
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Internal Staff AI

Must support:

- File Uploads
 - Images
 - Documents
 - Code Analysis
 - Knowledge Search
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Client AI

Restricted Mode.

Can access:

- Own Company Data
- Public Knowledge
- Support Assistance

Cannot access:

- Halo Internal Data
 - Other Clients
 - System Code
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AI Memory

Requirements:

- User Chat History
 - Compression
 - 10 Day Cleanup
 - Long Term Learning Storage
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AI Integration

AI available inside:

- Dashboard
 - Tickets
 - RMM
 - Security
 - Learning
 - HR
 - Accounting
 - Dev Console
 - Knowledge Base
-

AI Governance

Super Admin Override.

Owner access to all AI functions.

AI Operational Rights

Owner AI Mode:

Can request:

- Configuration Changes
- Feature Additions
- Report Generation
- Analysis

All actions must be audited.

END OF FILE 1 NEXT FILE: RMM + BACKUP + SECURITY + VAULT + SLA + CLIENT PORTAL